# Prabhavit Gupta

AI/ML Engineer: Production Agentic Workflows, Spec-Driven Systems & Evals

+39 351 610 7807 | prabhavitg@gmail.com | linkedin.com/in/prabhavit | github.com/prab-gupta | Venice, Italy (Remote EU / Freelance B2B Ready)

## Summary

AI/ML Engineer with 3 years of commercial production experience designing, operating, and evaluating production-grade Generative AI applications and agentic workflows. Experienced in architecting shared AI services, retrieval-augmented generation (RAG) pipelines, and multi-step tool-calling agents with deterministic validation guards. Emphasizes evaluation-driven engineering—implementing offline eval harnesses, golden benchmark datasets, and latency/cost monitoring. Skilled in Python backend services, relational data engineering (PostgreSQL), and cloud deployments on AWS and Linux. Fully authorized for freelance/B2B engagements in the European Union (Partita IVA registered).

## Experience

**FlowSpace.co** • *Lead Developer & AI Systems Architect (Freelance)*
Mar 2024 - Aug 2026 • London, UK (remote)

- Engineered production AI services and autonomous multi-step agent workflows automating unstructured document extraction (invoices, forms, identity records) for 4,500+ active enterprise users.
- Implemented structured tool-calling interfaces, dynamic retrieval layers, and Pydantic schema validation guards to eliminate hallucinated attributes and enforce downstream contracts.
- Designed and operated evaluation-driven loops using golden test datasets and LLM-as-a-judge scoring to benchmark model accuracy, grounding, and latency across prompt/model versions.
- Architected asynchronous task workers and queuing infrastructure with idempotency keys and retry policies, maintaining high service availability during heavy traffic surges.
- Optimized relational schemas and vector indexing in PostgreSQL for high-precision semantic search and structured entity retrieval.
- Managed automated CI/CD deployment pipelines, unit/integration testing suites (pytest), and Docker containerized microservices on AWS and Linux.
- *Tools: Python, PostgreSQL, Hugging Face, OpenAI/Anthropic APIs, Docker, AWS, Supabase, Linux*

**Urlink** • *Project Manager & Solutions Developer (Freelance)*
Apr 2025 - Jul 2025 • Barcelona, Spain (remote)

- Developed automated Python data extraction scripts and entity transformation pipelines connecting external APIs with central PostgreSQL repositories.
- Designed automated schema validation rules and data integrity checks, ensuring reliable cross-system synchronization for enterprise clients.
- Collaborated with cross-functional stakeholders to define interface specifications, SLA metrics, and operational monitoring alerts.
- *Tools: Python, REST APIs, PostgreSQL, Supabase, TypeScript, Linux*

**Independent Technical Consultant** • *AI Automation & Integration Developer*
Sep 2023 - Mar 2024 • Venice, Italy

- Built custom Python automation workers, webhook processors, and LLM extraction tools for automated data classification and database ingestion.
- Implemented offline test harnesses and synthetic test-case benchmarks to measure model output stability and detect schema regressions early.
- Authored detailed system documentation, error-handling runbooks, and API contracts for client technical teams.
- *Tools: Python, Webhooks, REST APIs, Supabase, Linux*

**Tractionboard.co** • *Low-Code Developer (Internship)*
Aug 2023 - Oct 2023 • Utrecht, Netherlands (remote)

- Integrated machine learning endpoints and notification services, establishing automated validation checks across core onboarding flows.
- *Tools: Python, OpenAI API, Sendgrid, REST APIs*

## Education

**Ca' Foscari University of Venice** • *Bachelor's in Philosophy, International Studies & Economics, Minor in Entrepreneurship*
Oct 2020 - Jun 2024 • Venice, Italy

## Certifications

Microsoft: Introduction to building AI agents • Microsoft Copilot Studio Agent Academy • Microsoft Power Automate • Tray.AI Build Practitioner (Sep 2024) • Tray.AI Build Foundations

## Additional Information

- **Work authorization:** European Long Term Residence Permit (Italy), unrestricted EU work authorization (Partita IVA / Italian freelance B2B ready). No visa sponsorship required.
- **Languages:** English (fluent / C1 professional), Hindi (native), Italian (working beginner).
- **Availability:** Available immediately for freelance / B2B remote roles in Europe.

## Core Skills

- **AI Engineering & Agent Systems:** Multi-step agent orchestration, tool calling, prompt engineering, RAG, embeddings, vector similarity search, Hugging Face, LLM APIs.
- **Evaluation & Reliability:** Offline eval harnesses, golden test datasets, LLM-as-a-judge pipelines, regression testing, drift detection, schema validation.
- **Backend & Data Systems:** Python 3, asynchronous processing, PostgreSQL schema design, indexing, SQL query optimization, Redis, REST APIs.
- **Cloud & DevOps:** Docker containerization, AWS, Linux system administration, Git, GitHub Actions CI/CD, task queues, logging and telemetry.
- **Engineering Practices:** Spec-driven development, clean architecture, automated testing (pytest), cross-functional technical communication.